

RAGatar: Enhancing LLM-driven Avatars with RAG for Knowledge-Adaptive Conversations in Virtual Reality

Alexander Marquardt*
Bonn-Rhein-Sieg University
of Applied Sciences
VAGO Solutions

David Golchinfar†
Bonn-Rhein-Sieg University
of Applied Sciences
VAGO Solutions

Daryoush Vaziri‡
Bonn-Rhein-Sieg University
of Applied Sciences
VAGO Solutions

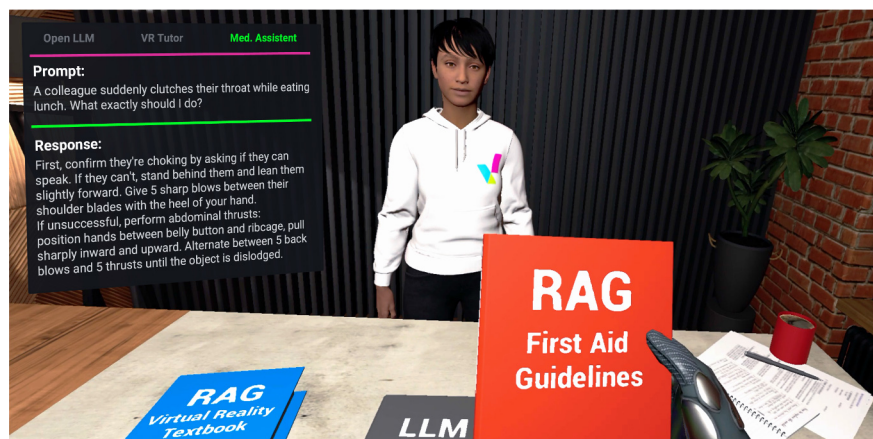


Figure 1: Real-time natural language conversation with an AI-driven avatar. Users can interact with different books in the virtual environment to control the avatar's knowledge domain, switching between general conversation and specialized expertise.

ABSTRACT

We present a virtual reality system that enables users to seamlessly switch between general conversations and domain-specific knowledge retrieval through natural interactions with AI-driven avatars. By combining MetaHuman technology with self-hosted large language models and retrieval-augmented generation, our system demonstrates how immersive AI interactions can enhance learning and training applications where both general communication and expert knowledge are required.

Index Terms: Virtual reality, artificial intelligence, interaction.

1 INTRODUCTION

The integration of virtual reality (VR) and artificial intelligence (AI) is transforming digital interactions. Generative AI plays a crucial role in the metaverse, enhancing interactions by enabling realistic and dynamic communication between users and AI-driven avatars [1]. Recent research has explored the potential of LLMs in enhancing VR environments. Ghosh et al. demonstrated virtual humans for clinical trial recruitment using text-based chat interaction in a desktop environment [2]. Wan et al. developed LLM-based AI agents in social VR, focusing on context-relevant responses with memory to simulate coherent interactions across sessions [4]. Lim et al. focused on conversational agents for health dialogues, emphasizing embodiment in user engagement [3].

However, these approaches are limited to single knowledge domains, do not fully leverage immersive VR experiences, and mostly rely on proprietary cloud-based services like OpenAI's ChatGPT.

This paper introduces a VR system that combines Unreal Engine's MetaHuman technology with a large language model (LLM), enabling natural language conversations with a multi-purpose avatar through a head-mounted display (HMD). Our system allows to seamlessly switch between general LLM access and specialized knowledge retrieval using retrieval-augmented generation (RAG). RAG enriches the system by linking the LLM to external resources for context-specific information beyond its initial training data. This approach enhances user experience by maintaining consistent interaction patterns while adapting its knowledge and expertise to different contexts, allowing the AI-driven avatar to serve as both a general conversation partner and a domain expert. By using self-hosted, open-source models, our architecture ensures full control over data privacy, allows fine-tuning for specific domains, and eliminates dependency on proprietary cloud services.

By combining immersive VR interaction with seamless switching between general and specialized knowledge through RAG, our research demonstrates how AI-driven avatars can deliver both flexible and context-specific interactions in virtual environments.

2 SYSTEM DESIGN

Our VR system enables natural language conversations with knowledge-adaptive avatars. We used a PC-VR setup (Intel Core i7-12700, 32 GB RAM, NVIDIA RTX 3070) with a Meta Quest 3 HMD. The system was developed in Unreal Engine 5.3.2 and employs the OpenXR API to enable VR capabilities and hand tracking with controllers, allowing users to engage intuitively with the virtual environment. Figure 2 illustrates the system's components and their interactions.

* e-mail: alexander.marquardt@h-brs.de

† e-mail: david.golchinfar@h-brs.de

‡ e-mail: daryoush.vaziri@h-brs.de

2.1 Virtual Environment and Avatar Behavior

The virtual environment is designed as a loft-office space featuring a custom female avatar created with MetaHuman Creator. The avatar exhibits animation states corresponding to its interaction status: idle, listening, thinking, and talking. These states are implemented using motion-capture animations for body and facial expressions, ensuring appropriate transitions during user interactions.

2.2 Interaction and Endpoint Selection

The system implements a selection mechanism through interactive books placed on a table in the virtual environment. Each book represents a specific endpoint – a dedicated processing pipeline that handles user queries through either a general LLM or specialized RAG implementation. We employ three distinct endpoints:

- **A general-purpose LLM for open-ended inquiries:**
This endpoint enables open-ended conversations, allowing users to engage in natural dialogue without domain constraints. Users can explore various topics and receive responses based on the LLM’s general knowledge base.
- **A RAG-enhanced endpoint for educational content:**
The educational endpoint integrates content from LaValle’s “Virtual Reality” textbook¹. This knowledge base enables precise answers to course-specific questions while maintaining the benefits of immersive interaction.
- **A RAG-enhanced endpoint for first aid training:**
This endpoint integrates emergency response procedures from the IFRC International First Aid Guidelines² through RAG, enabling the avatar to provide guidance for emergency scenarios. This implementation focuses on knowledge transfer and decision-making in critical situations.

Users interact with these endpoints through hand tracking enabled by the Meta Quest 3 controllers. The interaction process involves two steps: first, users select an endpoint by grabbing a book when in close proximity using the controller’s trigger button. Second, they initiate speech input by pressing and holding the controller’s A button, releasing it to complete their request. This design allows for seamless switching between different conversational contexts during runtime.

2.3 Natural Language Processing Pipeline

Our system processes natural dialogue using an open-source pipeline of language technologies. Speech input is first converted to text³, then directed to the appropriate endpoint – either the general LLM⁴ or a RAG-enhanced variant. For RAG-based interactions, the system employs embedding⁵ and reranking⁶ mechanisms to identify relevant information from the knowledge base. The selected endpoint then processes this input to generate a text response.

2.4 Response Generation and Avatar Animation

Once generated, the response is converted to speech using a multilingual text-to-speech service⁷. The audio output is processed by the OVR Lipsync plugin⁸, which creates 15 visemes to synchronize the avatar’s lip movements with the speech output. Together with the previously described animation states, this creates a cohesive interaction experience through the MetaHuman avatar.

¹<https://lavalle.pl/vr/vrbook4.pdf>

²https://www.ifrc.org/sites/default/files/2022-02/EN_GFARC_GUIDELINES_2020.pdf

³<https://huggingface.co/Systran/faster-whisper-large-v3>

⁴<https://huggingface.co/VAGOSolutions/SauerkrautLM-Nemo-12b-Instruct>

⁵<https://huggingface.co/intfloat/multilingual-e5-large>

⁶<https://huggingface.co/BAAL/bge-reranker-v2-m3>

⁷<https://huggingface.co/coqui/XTTS-v2>

⁸<https://developers.meta.com/horizon/documentation/unreal/audio-ovrlipsync/>

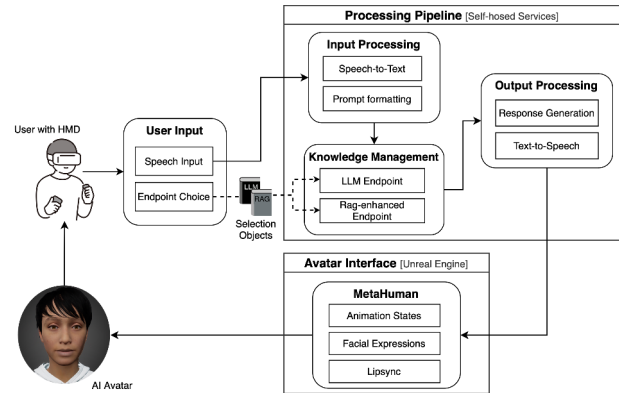


Figure 2: System architecture illustrating interaction and processing flow between user and AI-driven avatar.

3 CONCLUSION

Our research demonstrates how immersive AI interactions can be enhanced by combining flexible knowledge access and intuitive interaction design. By integrating RAG capabilities with general LLM functionality, we enable seamless switching between broad and specialized knowledge domains in VR-based AI interactions. This allows an AI-driven avatar to extend beyond general conversation capabilities by dynamically accessing domain-specific knowledge when needed. Such flexibility proves particularly valuable in scenarios requiring both general dialogue and specialized expertise, such as educational and professional training applications.

The system’s architecture offers several key advantages. By self-hosting all components, we maintain control over data privacy and customization, avoiding vendor lock-in and reducing costs associated with proprietary platforms like OpenAI’s ChatGPT. This approach allows for easy modification or fine-tuning of the language model to meet specific institutional needs.

Future work could explore automatic knowledge domain selection, determining appropriate expertise through techniques like intent recognition or function calling. This could eliminate the need for explicit domain selection while maintaining the benefits of specialized knowledge access. Furthermore, investigating multi-user scenarios and collaborative knowledge sharing could expand the system’s applicability in educational and professional settings.

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REFERENCES

- [1] V. Chamola, G. Bansal, T. K. Das, V. Hassija, S. Sai, J. Wang, S. Zeadally, A. Hussain, F. R. Yu, M. Guizani, and D. Niyato. Beyond reality: The pivotal role of generative ai in the metaverse. *IEEE Internet of Things Magazine*, 7(4):126–135, 2024. doi: 10.1109/IOTM.001.2300174 1
- [2] R. Ghosh, A. Maxim, C. You, and B. Lok. Real-time virtual human for promoting clinical trial education and recruitment. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1184–1185. IEEE, 2024. 1
- [3] S. Lim, R. Schmälzle, and G. Bente. Artificial social influence via human-embodied ai agent interaction in immersive virtual reality (vr): Effects of similarity-matching during health conversations. *arXiv preprint arXiv:2406.05486*, 2024. 1
- [4] H. Wan, J. Zhang, A. A. Suria, B. Yao, D. Wang, Y. Coady, and M. Prpa. Building llm-based ai agents in social virtual reality. In *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems, CHI EA ’24*. Association for Computing Machinery, New York, NY, USA, 2024. 1